

Elastic Load Balancing Task

Assignment Overview

This document covers six independent tasks: configuring a Classic Load Balancer for legacy HTTP routing, configuring an Application Load Balancer (ALB) for Layer 7 routing, configuring a Network Load Balancer (NLB) for high-performance Layer 4 routing, attaching an SSL/TLS certificate to the ALB, mapping the ALB to a custom domain in Route 53, and enabling ALB access log delivery to S3. Each task resets and is documented with its own independent steps.

Task	Resource	Purpose
1	Classic Load Balancer + EC2 instances	Legacy Layer 4/7 routing to backend instances
2	Application Load Balancer + Target Group	Layer 7 HTTP/HTTPS routing across instances
3	Network Load Balancer + Target Group	High-performance Layer 4 TCP routing, static IP
4	ACM Certificate + ALB HTTPS Listener	Encrypt traffic in transit via SSL/TLS
5	Route 53 Alias Record + ALB	Resolve a custom domain name to the ALB
6	ALB Attribute (Access Logs) + S3 Bucket	Store and analyze ALB access logs

Task 1: Configure a Classic Load Balancer

Outcome

A Classic Load Balancer named classic-Lb is active and distributing incoming HTTP traffic across two or more EC2 instances in different Availability Zones, with health checks confirming instance status.

Objective

To provide basic Layer 4/Layer 7 load balancing for a legacy application tier that does not require advanced routing rules, using the original Elastic Load Balancing offering.

Prerequisites

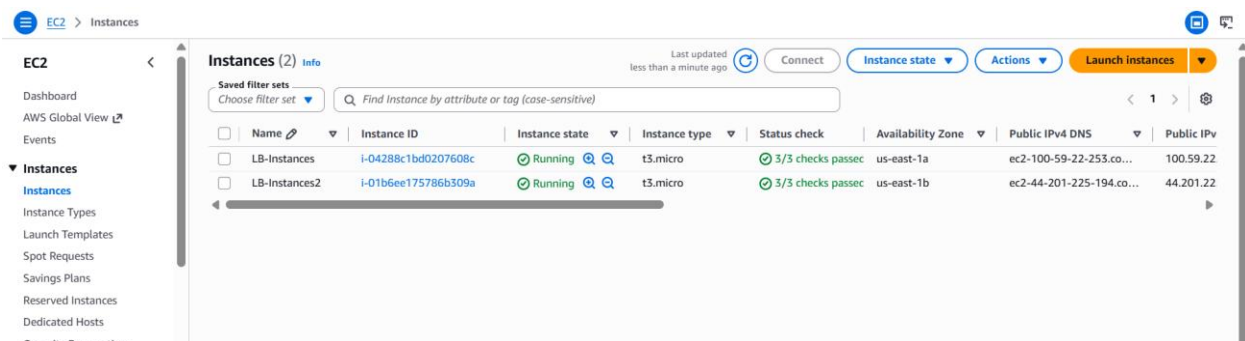
- Two or more running EC2 instances with a web server (e.g., Apache/Nginx) listening on port 80.
- A VPC with at least two subnets in different Availability Zones.
- A security group allowing inbound HTTP (port 80) traffic.

Step-by-Step Implementation

Step 1: Prepare Backend Instances

EC2 Console -> Instances -> confirm two instances are running and serving content on port 80. Attach a security group that allows inbound HTTP from the load balancer.

Screenshot Evidence:



Step 2: Create the Classic Load Balancer

EC2 Console -> Load Balancers -> Create Load Balancer -> Classic Load Balancer -> configure:

Parameter	Value
Load balancer name	classic-LB
Load balancer protocol/port	HTTP : 80
Instance protocol/port	HTTP : 80
VPC	Default or custom VPC
Availability Zones	At least two subnets selected

Screenshot Evidence

Screenshot Evidence

Health checks [Info](#)
Your load balancer automatically performs health checks to test the availability of all registered instances. Traffic is only routed to healthy instances, which is determined on their response to the health check.

Ping target
The health check ping is sent using the protocol and port you specify. If using HTTP/HTTPS protocol, you must also provide the destination path.

Ping protocol
HTTP

Ping path
/index.html

Ping port
80
1-65535

► **Advanced health check settings**

Instances (2) Remove Add instances
You can add instances to register as targets of the load balancer. Alternatively, after your load balancer is created, you can add it to an Amazon EC2 Auto Scaling group to ensure you maintain the correct number of instances to handle the load for your application. For maximum fault tolerance, we recommend maintaining approximately equivalent numbers of instances in each Availability Zone.

☐	Instance ID	Name	State	Security groups	Zone	Public IPv4 address	Subnet ID
☐	i-04288c1bd0207608c	LB-Instances	Running	LB-SecurityGroup	us-east-1a	100.59.22.253	subnet-06c0ff...
☐	i-01b6ee175786b309a	LB-Instances2	Running	LB-SecurityGroup	us-east-1b	44.201.225.194	subnet-0c7b1...

Step 4: Register Instances

Add the two prepared EC2 instances to the Classic Load Balancer and complete the wizard.

Screenshot Evidence:

Classic-LB Launch migration wizard Actions

▼ Details

Load balancer type
Classic

Status
0 of 2 instances in service

VPC
vpc-0a701668a4ff9329c

Date created
July 31, 2026, 13:03 (UTC+05:30)

Scheme
Internet-facing

Hosted zone
Z355XDOTRQ7X7K

Availability Zones
subnet-06c0ff69df529bbfe4 us-east-1a (use1-az1)
subnet-0c7b1347a08f35991 us-east-1b (use1-az2)

DNS name [Info](#)
Classic-LB-1974413197.us-east-1.elb.amazonaws.com (A Record)

► **Distribution of targets by Availability Zone (AZ)**
For each enabled Availability Zone, you can view the number of registered instances and their current health states. Selecting any values here will apply the corresponding filter to the Target instances table.

Listeners Network mapping Security Health checks Target instances Monitoring Attributes Tags

Listeners Manage listeners
A Classic Load Balancer listener uses the protocols and ports you configure, to both check for connection requests and forward received traffic to instances. The listener uses its protocol and port to check for connection requests. When traffic is received by the listener, it's forwarded to registered EC2 instances using the instance protocol and port.

Protocol:Port	Instance Pr...	Security policy	Default SSL/TLS certificate	Cookie stickiness	Backend server description
HTTP:80	HTTP:80	Not applicable	Not applicable	Not applicable	Not applicable

Validation Steps

- Both instances show InService in the Instances tab of the load balancer.
- The load balancer's public DNS name returns content from the web server when opened in a browser.
- Repeated requests are distributed across both registered instances.

Screenshot Evidence

[Insert screenshot: Classic Load Balancer DNS name serving traffic]



Apache HTTP Server Installed Successfully

Amazon Linux 2025

Hostname: ip-172-31-87-164.ec2.internal

Instance ID:

Private IP: 172.31.87.164



Apache HTTP Server Installed Successfully

Amazon Linux 2025

Hostname: ip-172-31-6-157.ec2.internal

Instance ID:

Private IP: 172.31.6.157

Task 2: Configure an Application Load Balancer

Outcome

An Application Load Balancer named app-lb-demo is active, forwarding HTTP traffic on port 80 to a target group containing healthy EC2 instances, with support for path- and host-based routing.

Objective

To provide Layer 7 load balancing with content-based routing, improved health checking, and native integration with target groups for a modern web application tier.

Prerequisites

- Two or more EC2 instances running a web server on port 80.
- A VPC with at least two public subnets in different Availability Zones.
- A security group allowing inbound HTTP (port 80) traffic.

Step-by-Step Implementation

Step 1: Create the Target Group

EC2 Console -> Target Groups -> Create target group -> configure:

Parameter	Value
Target type	Instances
Target group name	LB-TargetGroup
Protocol/port	HTTP : 80
VPC	Same VPC as EC2 instances
Health check path	/

Step 2: Register Targets

Register the EC2 instances to LB-TargetGroup and confirm both instances pass the health check.

Screenshot Evidence

[Insert screenshot: Registered targets showing healthy status]

The screenshot displays the AWS Management Console for an Elastic Load Balancing Target Group. The 'Details' section shows the target type as 'Instance', protocol as 'HTTP', and port as '80'. The 'Registered targets' table lists two targets, both with a 'Healthy' status.

Instance ID	Name	Port	Zone	Health status	Health status details	Admini...	Overrid...	Launch...	Anomaly detectio...
i-01b6ee175786b309a	LB-Instances2	80	us-east-1b (us...)	Healthy	Target group is not co...	-	-	July 31, 2...	Normal
i-04288c1bd0207608c	LB-Instances	80	us-east-1a (use...)	Healthy	Target group is not co...	-	-	July 31, 2...	Normal

Step 3: Create the Application Load Balancer

EC2 Console -> Load Balancers -> Create Load Balancer -> Application Load Balancer -> configure:

Parameter	Value
Load balancer name	My-ALB
Scheme	Internet-facing
IP address type	IPv4
VPC / Availability Zones	At least two public subnets selected
Security group	Allow inbound HTTP 80
Listener	HTTP : 80 -> forward to My-ALB

Screenshot Evidence

[Insert screenshot: Application Load Balancer configuration summary]

EC2 > Load balancers > Create Application Load Balancer

Create Application Load Balancer info

The Application Load Balancer distributes incoming HTTP and HTTPS traffic across multiple targets such as Amazon EC2 instances, microservices, and containers, based on request attributes. When the load balancer receives a connection request, it evaluates the listener rules in priority order to determine which rule to apply, and if applicable, it selects a target from the target group for the rule action.

► How Application Load Balancers work

Basic configuration

Load balancer name

Name must be unique within your AWS account and can't be changed after the load balancer is created.

My-ALB

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme info

Scheme can't be changed after the load balancer is created.

☒ Internet-facing

- Serves internet-facing traffic.
- Has public IP addresses.
- DNS name resolves to public IPs.
- Requires a public subnet.

☐ Internal

- Serves internal traffic.
- Has private IP addresses.
- DNS name resolves to private IPs.
- Compatible with the IPv4 and Dualstack IP address types.

Load balancer IP address type info

Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 addresses have an additional cost.

☒ IPv4

- Includes only IPv4 addresses.

☐ Dualstack

- Includes IPv4 and IPv6 addresses.

☐ Dualstack without public IPv4

- Includes a public IPv6 address, and private IPv4 and IPv6 addresses. Compatible with internet-facing load balancers only.

EC2 > Load balancers > Create Application Load Balancer

Network mapping info

The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC info

The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view target groups.

vpc-0a701668a4f9329c172.31.0.0/16 (default) Create VPC

IP pools info

You can optionally choose to configure an IPAM pool as the preferred source for your load balancers IP addresses. Create or view Pools in the Amazon VPC IP Address Manager console.

☐ Use IPAM pool for public IPv4 addresses

- The IPAM pool you choose will be the preferred source of public IPv4 addresses. If the pool is depleted IPv4 addresses will be assigned by AWS.

Availability Zones and subnets info

Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.

☒ us-east-1a (use1-az1)Subnet

- Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently. Selected subnets must have Network ACL inbound and outbound ALLOW rules, and internet gateway (IGW) routes configured to allow traffic.

subnet-06c0f69d5f529bbfe4172.31.0.0/20Network ACL ALLOW rules: Inbound (1) Outbound (1) | Route table: IGW (-/)

☒ us-east-1b (use1-az2)Subnet

- Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently. Selected subnets must have Network ACL inbound and outbound ALLOW rules, and internet gateway (IGW) routes configured to allow traffic.

subnet-0c7b1347a08f55991172.31.0.0/20Network ACL ALLOW rules: Inbound (1) Outbound (1) | Route table: IGW (-/)

☐ us-east-1c (use1-az4)

Security groups info

A security group is a set of firewall rules that control the traffic to your load balancer. Listed security groups are in the same VPC as you selected above. Selected security groups must include at least 1 inbound rule and 1 outbound rule to allow traffic to your load balancer.

Security groups

Select up to 5 security groups

default (sg-094cb17dff32b9d7f) Inbound rule (1) Outbound rule (1) Create security group

Listeners and routing info

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

▼ Listener HTTP80 Remove

Protocol HTTP Port 801-65535

Default action info

The default action is used if no other rules apply. Choose the default action for traffic on this listener.

Routing action

☒ Forward to target groups☐ Redirect to URL☐ Return fixed response

Forward to target group info

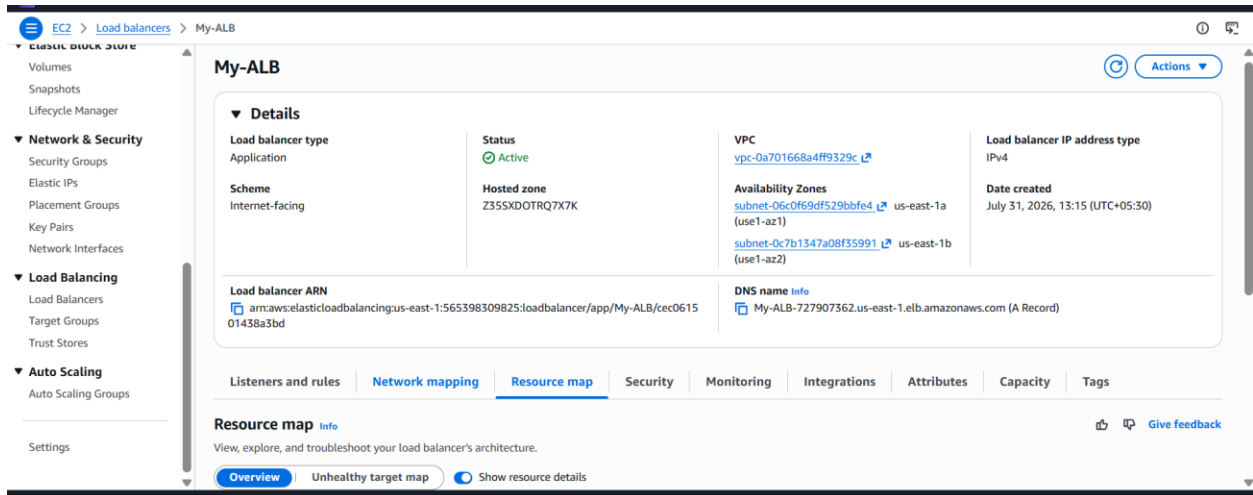
Choose a target group and specify routing weight or create target group.

Target group

LB-TargetGroup HTTP Target type: Instance, IPv4 | Target stickiness: Off

Weight 10-999

Percent 100%



Validation Steps

- The target group shows both instances as healthy.
- The ALB's DNS name serves the web application over HTTP.
- Traffic alternates between backend instances on repeated requests.

Screenshot Evidence

[Insert screenshot: ALB DNS name responding successfully in browser]



Apache HTTP Server Installed Successfully

Amazon Linux 2025

Hostname: ip-172-31-6-157.ec2.internal

Instance ID:

Private IP: 172.31.6.157



Apache HTTP Server Installed Successfully

Amazon Linux 2025

Hostname: ip-172-31-87-164.ec2.internal

Instance ID:

Private IP: 172.31.87.164

Task 3: Configure a Network Load Balancer

Outcome

A Network Load Balancer named network-Lb is active, forwarding TCP traffic on port 80 to a target group of EC2 instances, with a static IP address available per Availability Zone.

Objective

To provide ultra-low-latency Layer 4 load balancing for TCP/UDP workloads that require static IP addresses or extreme throughput, such as high-volume APIs or non-HTTP protocols.

Prerequisites

- Two or more EC2 instances accepting TCP connections on port 80.
- A VPC with at least two subnets in different Availability Zones.
- A security group allowing inbound TCP traffic on port 80.

Step-by-Step Implementation

Step 1: Create the Target Group

EC2 Console -> Target Groups -> Create target group -> configure:

Parameter	Value
Target type	Instances
Target group name	NLB-TargetGroup
Protocol/port	TCP : 80
VPC	Same VPC as EC2 instances
Health check protocol	TCP

Screenshot Evidence

[Insert screenshot: TCP target group creation]

Settings - immutable

Choose a target type and the load balancer and listener will route traffic to your target. These settings can't be modified after target group creation."

Target type

Indicate what resource type you want to target. Only the selected resource type can be registered to this target group.

☒ Instances
 Supports load balancing to instances in a VPC. Integrate with Auto Scaling Groups or ECS services for automatic management.
 Suitable for: ALB NLB GWLB

☐ IP addresses
 Supports load balancing to VPC and on-premises resources. Facilitates routing to IP addresses and network interfaces on the same instance. Supports IPv6 targets.
 Suitable for: ALB NLB GWLB

☐ Lambda function
 Supports load balancing to a single Lambda function. ALB required as traffic source.
 Suitable for: ALB

☐ Application Load Balancer
 Allows use of static IP addresses and PrivateLink with an Application Load Balancer. NLB required as traffic source.
 Suitable for: NLB

Target group name

Name must be unique per Region per AWS account.

NLB-TargetGroup

Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-32 total characters; Count: 15/32

Protocol

Protocol for communication between the load balancer and targets.

TCP

Port

Port number where targets receive traffic. Can be overridden for individual targets during registration.

80

1-65535

IP address type

Only targets with the indicated IP address type can be registered to this target group.

☒ IPv4
 Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.

Step 2: Register Targets

Register the EC2 instances to NLB-TargetGroup and confirm healthy status.

Screenshot Evidence

[Insert screenshot: Healthy targets registered to the NLB target group]

NLB-TargetGroup									
Details									
Target type									
Instance									
Load balancer									
None associated									
Total targets									
2									
Healthy									
0									
Unhealthy									
0									
Unused									
2									
Initial									
0									
Draining									
0									
Distribution of targets by Availability Zone (AZ)									
Select values in this table to see corresponding filters applied to the Registered targets table below.									
Targets									
Monitoring									
Health checks									
Attributes									
Tags									
Registered targets (2)									
Filter targets									
Deregister									
Register targets									
Instance ID									
Name									
Port									
Zone									
Health status									
Health status details									
Administrative ...									
Override details									
Launch time									
i-01b6ee175786b3...									
LB-Instances2									
80									
us-east-1b (use1-a...									
Unused									
Target group is not...									
-									
July 31, 2026, 12:5...									
i-04288c1bd02076...									
LB-Instances									
80									
us-east-1a (use1-az1)									
Unused									
Target group is not...									
-									
July 31, 2026, 12:5...									

Step 3: Create the Network Load Balancer

EC2 Console -> Load Balancers -> Create Load Balancer -> Network Load Balancer -> configure:

Parameter	Value
Load balancer name	My-NLB
Scheme	Internet-facing

Parameter	Value
Availability Zones	At least two subnets, one Elastic IP per zone
Listener	TCP : 80 -> forward to NLB-TargetGroup

Screenshot Evidence

[Insert screenshot: Network Load Balancer listener and Elastic IP assignment]

EC2 > Load balancers > Create Network Load Balancer

Create Network Load Balancer

The Network Load Balancer distributes incoming TCP and UDP traffic across multiple targets such as Amazon EC2 instances, microservices, and containers. When the load balancer receives a connection request, it selects a target based on the protocol and port that are specified in the listener configuration, and the routing rule specified as the default action.

How Network Load Balancers work

Basic configuration

Load balancer name

Name must be unique within your AWS account and can't be changed after the load balancer is created.

My-NLB

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme

Scheme can't be changed after the load balancer is created.

Internet-facing

- Serves internet-facing traffic.
- Has public IP addresses.
- DNS name resolves to public IPs.
- Requires a public subnet.

Internal

- Serves internal traffic.
- Has private IP addresses.
- DNS name resolves to private IPs.

Load balancer IP address type

Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types.

IPv4

Includes only IPv4 addresses.

Dualstack

Includes IPv4 and IPv6 addresses.

Network mapping

The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC

The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to on-premises targets or if using VPC peering. To confirm the VPC for your targets, view [target groups](#).

vpc-0a701668a4ff9329c

172.31.0.0/16

(default)

Create VPC

Availability Zones and subnets

Select one or more Availability Zones and corresponding subnets. Enabling multiple Availability Zones increases the fault tolerance of your applications. The load balancer routes traffic to targets in the selected Availability Zones only. Availability Zones that are not supported by the load balancer or the VPC are not available for selection.

us-east-1a (use1-az1)

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently. Selected subnets must have Network ACL inbound and outbound ALLOW rules, and internet gateway (IGW) routes configured to allow traffic.

subnet-06c0f69df529bbfe4

172.31.0.0/20

Network ACL ALLOW rules: Inbound (1) Outbound (1) | Route table: IGW (-)

IPv4 address

The front-end IPv4 address of the load balancer in the selected Availability Zone.

Assigned by AWS

Use an Elastic IP address

us-east-1b (use1-az2)

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently. Selected subnets must have Network ACL inbound and outbound ALLOW rules, and internet gateway (IGW) routes configured to allow traffic.

subnet-0c7b1347a08f55991

172.31.80.0/20

Network ACL ALLOW rules: Inbound (1) Outbound (1) | Route table: IGW (-)

IPv4 address

The front-end IPv4 address of the load balancer in the selected Availability Zone.

Security groups info

A security group is a set of firewall rules that control the traffic to your load balancer. Listed security groups are in the same VPC as you selected above. Selected security groups must include at least 1 inbound rule and 1 outbound rule to allow traffic to your load balancer.

Security groups - recommended

Security groups support on Network Load Balancers can only be enabled at creation by including at least one security group. You can change security groups after creation. The security groups for your load balancer must allow it to communicate with registered targets on both the listener port and the health check port. For PrivateLink Network Load Balancers, security group rules are enforced on PrivateLink traffic; however, you can turn off inbound rule evaluation after creation within the load balancer's Security tab or using the API.

Select up to 5 security groups

mysg (sg-08c0ae7970e16fa6d) X

LB-SecurityGroup (sg-0d9d756dc72fde311) X

Inbound rule (1)

Outbound rule (1)

Inbound rule (2)

Outbound rule (1)

Create security group

Listeners and routing info

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

▼ Listener TCP:80

Remove

Protocol

TCP

Port

80

1-65535

Forward to target group info

Choose a target group and specify routing weight or [create target group](#).

Target group

NLB-TargetGroup

Target type: Instance, IPv4 | Target stickiness: Off

TCP

Weight

1

0-999

Percent

100%

Validation Steps

- Target group shows both instances as healthy.
- The NLB's static Elastic IP(s) or DNS name accept TCP connections on port 80.
- Connections succeed consistently with low latency.

Screenshot Evidence

[Insert screenshot: Successful TCP connection test against the NLB]

My-NLB

Actions

▼ Details

Load balancer type

Network

Scheme

Internet-facing

Status

Active

Hosted zone

Z26RNL4JYFTOTI

VPC

vpc-0a701668a4ff9329c

Availability Zones

subnet-06c0f69df529bbfe4 us-east-1a (use1-az1)

47a08f35991 us-east-1b (use1-az2)

Load balancer IP address type

IPv4

Date created

July 31, 2026, 14:50 (UTC+05:30)

Load balancer ARN

arn:aws:elasticloadbalancing:us-east-1:565398309825:loadbalancer/net/My-NLB/f925307527f572f2

DNS name copied

My-NLB-f925307527f572f2.elb.us-east-1.amazonaws.com (A Record)

Listeners and rules (1) info

Manage rules

Manage listener

Add listener

A listener checks for connection requests using the protocol and port that you configure. Traffic received by a Network Load Balancer listener is forwarded to the selected target group.

Filter listeners

Protocol:Port

Default action

Rules

ARN

Security policy

Default SSL/TLS certificate

Tags

TCP:80

Forward to target group

NLB-TargetGroup 1 (100%)

Target group stickiness: Off

1 rule

ARN

Not applicable

Not applicable

0 tags

Apache HTTP Server Installed Successfully

Amazon Linux 2025

Hostname: ip-172-31-6-157.ec2.internal

Instance ID:

Private IP: 172.31.6.157



Not secure

my-nlb-f925307527f572f2.elb.us-east-1.amazonaws.com



Apache HTTP Server Installed Successfully

Amazon Linux 2025

Hostname: ip-172-31-87-164.ec2.internal

Instance ID:

Private IP: 172.31.87.164

Task 4: Attach SSL/TLS Certificate to the Application Load Balancer

Outcome

The Application Load Balancer (app-lb-demo) has an HTTPS:443 listener secured with an AWS Certificate Manager (ACM) certificate, and HTTP:80 traffic is automatically redirected to HTTPS.

Objective

To encrypt traffic in transit between clients and the load balancer, satisfying security best practices and enabling HTTPS access to the application.

Prerequisites

- An existing Application Load Balancer (app-lb-demo) with a healthy target group.
- A registered domain name or subdomain to validate the certificate against.
- Access to DNS validation records (Route 53 or external DNS provider).

Step-by-Step Implementation

Step 1: Request a Certificate in ACM

Certificate Manager Console -> Request a certificate -> Request a public certificate -> configure:

Parameter	Value
Domain name	Krishna10.in
Validation method	DNS validation
Key algorithm	RSA 2048

Screenshot Evidence

[Insert screenshot: ACM certificate request and DNS validation records]

Details

In use No	Serial number N/A	Requested at July 31, 2026, 15:01:18 (UTC+5:30)	Renewal eligibility Ineligible
Domain name krishna10.in	Public key info RSA 2048	Issued at N/A	Export option Disabled
Number of additional names 0	Signature algorithm SHA-256 with RSA	Not before N/A	
	Can be used with CloudFront, Elastic Load Balancing, API Gateway and other integrated services. View details	Not after N/A	

Tags (0)
A tag is a label that you assign to an AWS resource.

Key	Value
No tags	
No tags associated with this resource.	

Manage tags

Step 2: Validate the Certificate

Add the CNAME validation record provided by ACM to Route 53 (or the domain's DNS provider) and wait for the certificate status to change to Issued.

Screenshot Evidence

[Insert screenshot: ACM certificate status showing Issued]

f4a2ded3-4440-461e-a71a-2c206661a044

Delete

Certificate status

Identifier

f4a2ded3-4440-461e-a71a-2c206661a044

ARN

arn:aws:acm:us-east-1:565398309825:certificate/f4a2ded3-4440-461e-a71a-2c206661a044

Type

Amazon Issued

Status

Issued

Key source

AWS managed

Domains (1)

Create records in Route 53

Export to CSV

< 1 >

Domain	Status	Renewal status	Type	CNAME name	CNAME value
krishna10.in	Success	-	CNAME	_250d3e8e8fdb4fd74eca2fd56338ac1b.krishna10.in.	_7d14d65d8d680550494bb88daa6cf489.validations.aws.

Details

In use

No

Serial number

01:c5:48:45:fc:2c:1f:c8:b2:0d:73:6f:3e:91:f9:62

Requested at

July 31, 2026, 15:31:14 (UTC+5:30)

Renewal eligibility

Ineligible

Domain name

krishna10.in

Public key info

RSA 2048

Issued at

Jul 31 2026 15:54:55 (UTC+5:30)

Export option

Disabled

Step 3: Add an HTTPS Listener to the ALB

EC2 Console -> Load Balancers -> My-ALB -> Listeners -> Add listener -> configure:

Parameter	Value
Protocol/port	HTTPS : 443
Default action	Forward to LB-TargetGroup
Security policy	ELBSecurityPolicy-TLS13-1-2-2021-06
Certificate source	ACM certificate for app.example.com

Screenshot Evidence

[Insert screenshot: HTTPS listener configuration with ACM certificate attached]

Listener: HTTPS:443

A listener checks for connection requests on its configured protocol and port. Incoming requests are evaluated against rules in priority order; the first matching rule determines routing. If no rule matches, the default action is applied.

Protocol

Used for connections from clients to the load balancer.

HTTPS

Port

The port on which the load balancer is listening for connections.

443

1-65535

Default action

The default action is used if no other rules apply. Choose the default action for traffic on this listener.

No pre-routing action

Authenticate user

Validate token

Routing action

Forward to target groups

Redirect to URL

Return fixed response

Forward to target group

Choose a target group and specify routing weight or create target group

LB-TargetGroup

Target type: Instance, IPv4 | Target stickiness: Off

HTTP

Weight

1

Percent

100%

Target group stickiness

Enables the load balancer to bind a user's session to a specific target group. To use stickiness the client must support cookies. If you want to bind a user's session to a specific target, turn on the Target Group attribute Stickiness.

Turn on target group stickiness

Secure listener settings info

Security policy info

Your load balancer uses a Secure Socket Layer (SSL) negotiation configuration called a security policy to manage SSL connections with clients. [Compare security policies](#)

Security category

All security policies

Policy name New

ELBSecurityPolicy-TLS13-1-2-Res-PQ-2025-09 (Recommended)

Default SSL/TLS server certificate

The certificate used if a client connects without SNI protocol, or if there are no matching certificates. You can source this certificate from AWS Certificate Manager (ACM), Amazon Identity and Access Management (IAM), or import a certificate. This certificate will automatically be added to your listener certificate list.

Certificate source

☒ From ACM

☐ From IAM

☐ Import certificate

Certificate (from ACM)

The selected certificate will be applied as the default SSL/TLS server certificate for this load balancer's secure listeners.

krishna10.in

f4a2ded3-4440-461e-a71a-2c206661a044

[Request new ACM certificate](#)

Client certificate handling info

Client certificates are used to make authenticated requests to remote servers. [Learn more](#)

☐ Mutual authentication (mTLS)

Mutual TLS (Transport Layer Security) authentication offers two-way peer authentication. It adds a layer of security over TLS and allows your services to verify the client that's making the connection.

Default attributes are applied when adding a listener. You can edit them after adding the listener.

Step 4: Redirect HTTP to HTTPS

Edit the existing HTTP:80 listener rule and change the default action to Redirect to HTTPS:443 (301 redirect).

Screenshot Evidence

[Insert screenshot: HTTP listener rule updated to redirect to HTTPS]

Listeners and rules	Network mapping	Resource map	Security	Monitoring	Integrations	Attributes	Capacity	Tags
Listeners and rules (2) info								
A listener checks for connection requests on its configured protocol and port. Incoming requests are evaluated against rules in priority order; the first matching rule determines routing. If no rule matches, the default action is applied.								
Filter listeners								
☐	Protocol:Port	Default action	Rules	ARN	Security policy	Default SSL/TLS certificate	mTLS	Trust sto
☐	HTTP:80	Forward to target group LB-TargetGroup 1: 1 (100%) Target group stickiness: Off	1 rule	ARN	Not applicable	Not applicable	Not applicable	Not appl
☐	HTTPS:443	Forward to target group LB-TargetGroup 1: 1 (100%) Target group stickiness: Off	1 rule	ARN	ELBSecurityPolicy-TLS13-1-2-...	krishna10.in (Certificate ID: f4...	Off	Not appl

Validation Steps

- Browsing to http://app.example.com automatically redirects to https://app.example.com.
- The browser shows a valid certificate (padlock icon) with no warnings.
- The ALB listener list shows both HTTP:80 (redirect) and HTTPS:443 (forward) listeners active.

Screenshot Evidence

[Insert screenshot: Browser padlock confirming a valid HTTPS connection]

Apache HTTP Server Installed Successfully

Amazon Linux 2025

Hostname: ip-172-31-87-164.ec2.internal

Instance ID:

Private IP: 172.31.87.164

Task 5: Map the Application Load Balancer to Route 53

Outcome

A custom domain name (app.example.com) resolves directly to the Application Load Balancer through a Route 53 Alias record, eliminating the need to expose the raw ALB DNS name to users.

Objective

To provide a friendly, brandable domain name for the application while retaining the automatic health-aware routing benefits of an Alias record over a standard CNAME.

Prerequisites

- A public hosted zone in Route 53 for the domain (e.g., example.com).
- An active Application Load Balancer (app-lb-demo), ideally with the HTTPS listener from Task 4 already configured.

Step-by-Step Implementation

Step 1: Open the Hosted Zone

Route 53 Console -> Hosted zones -> select example.com.

Step 2: Create the Alias Record

Create record -> configure:

Parameter	Value
Record name	Krishna10.in
Record type	A - Routes traffic to an IPv4 address
Alias	Yes
Route traffic to	Alias to Application and Classic Load Balancer
Region	Region where app-lb-demo is deployed
Load balancer	My-ALB

Screenshot Evidence

[Insert screenshot: Alias record configuration pointing to the ALB]

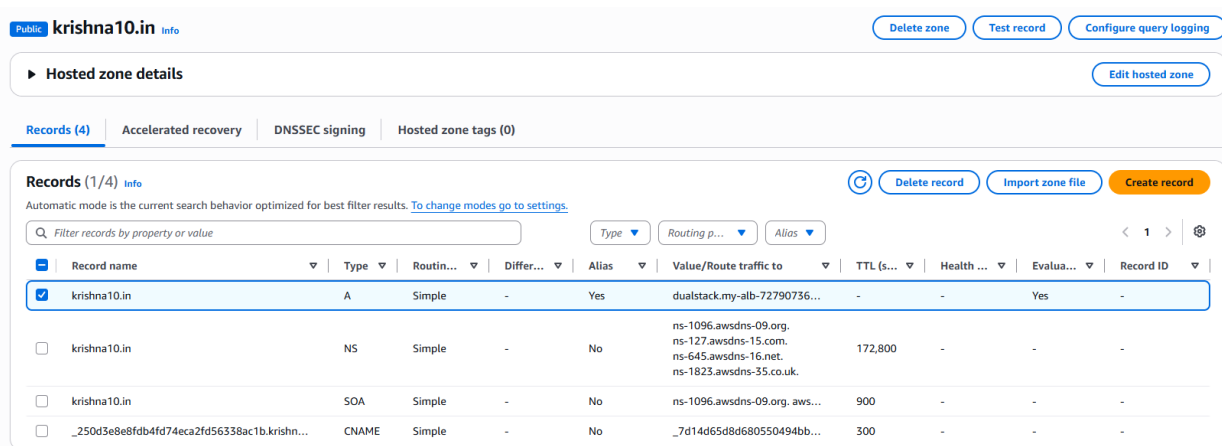
The screenshot shows the 'Create record' interface in the AWS Route 53 console. The 'Quick create record' section is active, showing a record named 'krishna10.in' of type 'A - Routes traffic to an IPv4 address and some AWS resources'. The 'Alias' checkbox is checked. The 'Route traffic to' dropdown is set to 'Alias to Application and Classic Load Balancer', and the 'Region' dropdown is set to 'US East (N. Virginia)'. The 'Load balancer' field is populated with 'dualstack.My-ALB-727907362.us-east-1.elb.amazonaws.com'. The 'Routing policy' is set to 'Simple routing', and 'Evaluate target health' is checked. The 'Add another record' button is visible at the bottom right.

Step 3: Save and Propagate

Create the record and allow DNS propagation, typically completing within a few minutes due to Route 53's low TTL for Alias records.

Screenshot Evidence

[Insert screenshot: Confirmation of the newly created A/Alias record]



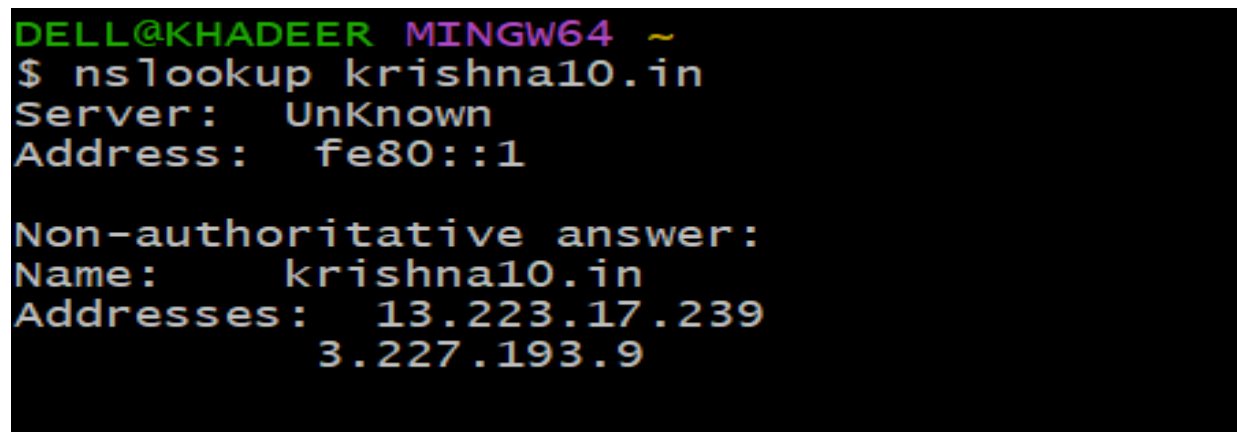
Record name	Type	Routin...	Differ...	Alias	Value/Route traffic to	TTL (s...)	Health ...	Evalu...	Record ID
☒ krishna10.in	A	Simple	-	Yes	dualstack.my-alb-72790736...	-	-	Yes	-
☐ krishna10.in	NS	Simple	-	No	ns-1096.awsdns-09.org, ns-127.awsdns-15.com, ns-645.awsdns-16.net, ns-1823.awsdns-35.co.uk.	172,800	-	-	-
☐ krishna10.in	SOA	Simple	-	No	ns-1096.awsdns-09.org. aws...	900	-	-	-
☐ _250d3e8e8fdb4fd74eca2fd56338ac1b.krishn...	CNAME	Simple	-	No	_7d14d65d8d680550494bb...	300	-	-	-

Validation Steps

- A DNS lookup (dig/nslookup) for app.example.com resolves to the ALB's IP addresses.
- Browsing to https://app.example.com loads the application without referencing the ALB DNS name directly.
- The record shows as an Alias in the Route 53 console, not a CNAME.

Screenshot Evidence

[Insert screenshot: dig/nslookup output resolving the custom domain]



```
DELL@KHADEER MINGW64 ~
$ nslookup krishna10.in
Server: UnKnown
Address: fe80::1

Non-authoritative answer:
Name: krishna10.in
Addresses: 13.223.17.239
           3.227.193.9
```

Apache HTTP Server Installed Successfully

Amazon Linux 2025

Hostname: ip-172-31-87-164.ec2.internal

Instance ID:

Private IP: 172.31.87.164

Task 6: Push Application Load Balancer Logs to S3

Outcome

Access logging is enabled on the Application Load Balancer (app-lb-demo), and detailed request-level logs are delivered as compressed files to a dedicated S3 bucket for traffic analysis and auditing.

Objective

To capture request-level access logs (client IP, latency, response codes, target processing time) for troubleshooting, security review, and long-term traffic analysis.

Prerequisites

- An existing Application Load Balancer (app-lb-demo).
- Permissions to create and configure an S3 bucket and bucket policy.

Step-by-Step Implementation

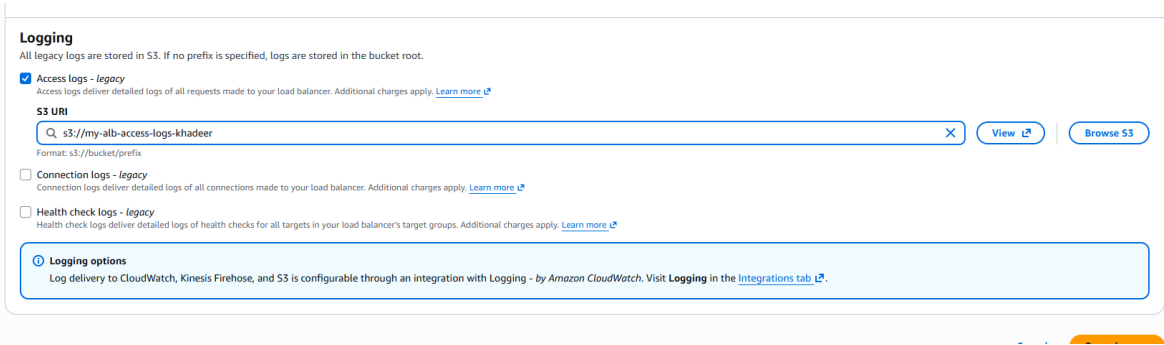
Step 1: Create the S3 Bucket for Access Logs

S3 Console -> Create bucket -> configure:

Parameter	Value
Bucket name	My-alb-access-logs-khadeer
Region	Same region as the ALB
Block Public Access	Enabled (all four settings)
Default encryption	SSE-S3

Screenshot Evidence

[Insert screenshot: S3 bucket created for ALB access logs]



Step 2: Attach the Required Bucket Policy

Attach a bucket policy granting the AWS load-balancing service account (specific to the ALB's region) permission to write objects under the chosen log prefix.

Screenshot Evidence

[Insert screenshot: Bucket policy granting ELB log delivery permissions]

Bucket policy

EditDelete

The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. [Learn more](#)

Public access is blocked because Block Public Access settings are turned on for this bucket

To determine which settings are turned on, check your Block Public Access settings for this bucket. Learn more about [using Amazon S3 Block Public Access](#)

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "AWSLogDeliveryWrite",
      "Effect": "Allow",
      "Principal": {
        "Service": "logdelivery.elasticloadbalancing.amazonaws.com"
      },
      "Action": "s3:PutObject",
      "Resource": "arn:aws:s3::my-alb-access-logs-khadeer/AWSLogs/565398309825/*"
    },
    {
      "Sid": "AWSLogDeliveryAclCheck",
      "Effect": "Allow",
      "Principal": {
        "Service": "logdelivery.elasticloadbalancing.amazonaws.com"
      },
      "Action": "s3:GetBucketAcl",
      "Resource": "arn:aws:s3::my-alb-access-logs-khadeer"
    }
  ]
}
```

Copy

Step 3: Enable Access Logs on the ALB

EC2 Console -> Load Balancers -> My-alb-access-logs-khadeer -> Attributes -> Edit -> configure:

Parameter	Value
Access logs	Enabled
S3 bucket	My-alb-access-logs-khadeer
S3 prefix	S3://bucket/perfix

Screenshot Evidence

[Insert screenshot: ALB attributes screen with access logging enabled]

Route 53

Amazon S3 > Buckets > my-alb-access-logs-khadeer > AWSLogs/

Amazon S3

Buckets

General purpose buckets

Directory buckets

Table buckets

Vector buckets

Files

File systems

Access management and security

Access Points

Access Points for FSx

Access Grants

IAM Access Analyzer

Storage management and lifecycle

AWSLogs/

ObjectsProperties

Objects (1)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 Inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

Show versions

NameTypeLast modifiedSizeStorage class

565398309825/

Folder-

Validation Steps

- New .log.gz objects appear under the alb-access-logs-demo bucket's AWSLogs/ prefix within a few minutes of traffic.
- Downloaded log files contain request-level fields such as client IP, target status code, and processing time.
- Log delivery continues consistently as new requests reach the ALB.

Screenshot Evidence

[Insert screenshot: S3 bucket showing delivered ALB access log objects]

- Amazon S3 <
- ▼ Buckets

General purpose buckets

Directory buckets

Table buckets

Vector buckets
- ▼ Files

File systems
- ▼ Access management and security

Access Points

Access Points for FSx

Access Grants

IAM Access Analyzer
- ▼ Storage management and insights

Storage Lens

Batch Operations
- Account and organization settings
- AWS Marketplace for S3

565398309825_elasticloadbalancing_us-east-1_app.My-ALB.7491e8b406ea245f_20260730T0820Z_98.91.121.114_e20s8mcm.log.gz Info

Copy S3 URI

Download

Open

Object actions

Properties | Permissions | Versions

Object overview

Owner

e9c7cd174088b21aafdc30a69f8f25bb5e807e5d1833b04bae49851d35a1e879c

AWS Region

US East (N. Virginia) us-east-1

Last modified

July 30, 2026, 13:50:11 (UTC+05:30)

Size

429.0 B

Type

gz

Key

AWSLogs/565398309825/elasticloadbalancing/us-east-1/2026/07/30/565398309825_elasticloadbalancing_us-east-1_app.My-ALB.7491e8b406ea245f_20260730T0820Z_98.91.121.114_e20s8mcm.log.gz

S3 URI

s3://my-alb-access-logs-khadeer/AWSLogs/565398309825/elasticloadbalancing/us-east-1/2026/07/30/565398309825_elasticloadbalancing_us-east-1_app.My-ALB.7491e8b406ea245f_20260730T0820Z_98.91.121.114_e20s8mcm.log.gz

Amazon Resource Name (ARN)

arn:aws:s3::my-alb-access-logs-khadeer/AWSLogs/565398309825/elasticloadbalancing/us-east-1/2026/07/30/565398309825_elasticloadbalancing_us-east-1_app.My-ALB.7491e8b406ea245f_20260730T0820Z_98.91.121.114_e20s8mcm.log.gz

Entity tag (ETag)

62a825a364a5ad1b99aee40e42ed66f5

Object URL

https://my-alb-access-logs-khadeer.s3.us-east-1.amazonaws.com/AWSLogs/565398309825/elasticloadbalancing/us-east-1/2026/07/30/565398309825_elasticloadbalancing_us-east-1_app.My-ALB.7491e8b406ea245f_20260730T0820Z_98.91.121.114_e20s8mcm.log.gz

Object management overview

The following bucket properties and object management configurations impact the behavior of this object.